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Specialized Topics of Theoretical Physics

Majorana Modes in the Machine –
Simulating Topological Phases with Quantum Circuits and AI

NISQ Noise on the Diagnostic

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SS 26

Contents

1 Introduction

The 1D Kitaev chain hosts unpaired Majorana zero modes at its ends in the topological phase [1].

2 Noise model and diagnostic

2.1 Gate-level quantum channels

$$\widetilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \qquad \mathcal{E}_g(\rho) = \sum_k K_k \rho K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I. \tag{2.1}$$

$$\mathcal{E}_p^{\mathrm{dep}}(\rho) = (1-p)\rho + p\,\frac{\operatorname{Tr}\rho}{2^n} I. \tag{2.2}$$

2.2 Three locations for error

$$\langle O_{\mathrm{edge}} \rangle_{\mathrm{readout}} = (1-2\epsilon)^L \langle O_{\mathrm{edge}} \rangle, \tag{2.3}$$

2.3 Expressibility versus accumulated noise

$$|\langle O_{\mathrm{edge}} \rangle|_{\mathrm{meas}}(r) \simeq |\langle O_{\mathrm{edge}} \rangle|_{\mathrm{ideal}}(r)(1-p_{\mathrm{cx}})^{r(L-1)}. \tag{2.4}$$

$$C_{\mathrm{noisy}}(\theta) = \operatorname{Tr}[H\rho_\theta] + \lambda\big(1 - \operatorname{Tr}[P\rho_\theta]\big)^2, \qquad \rho_\theta = \mathcal{E}\Big(U(\theta)|0\rangle\langle 0|U^\dagger(\theta)\Big). \tag{2.5}$$

3 Methods and Results

3.1 Circuit-level implementation and verification

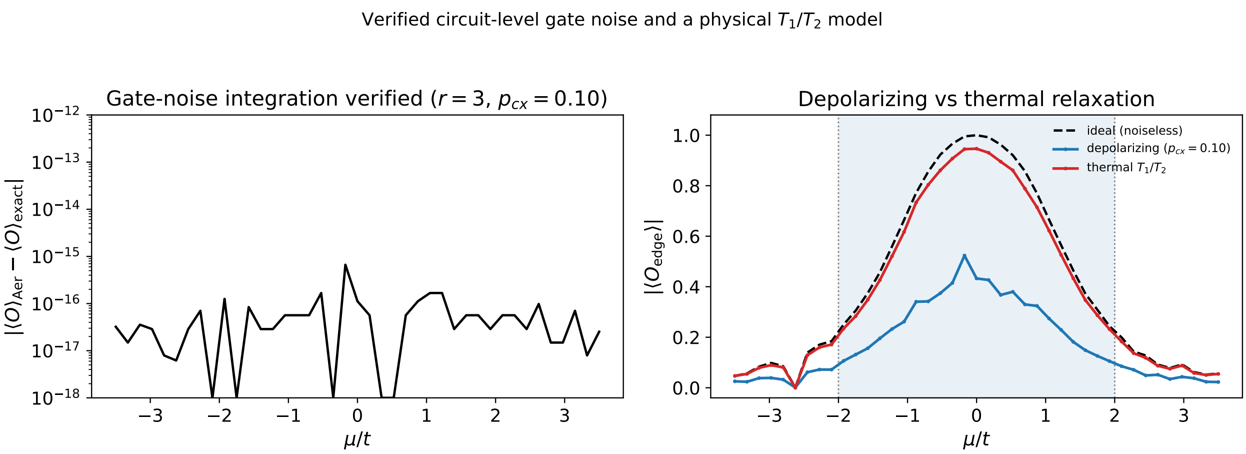

Figure 1: Left: the Aer `NoiseModel` edge string and the independent exact gate-by-gate reference agree to $\sim 10^{-15}$ across the μ sweep. Right: depolarizing versus thermal T_1/T_2 noise at matched gate placement.

3.2 The optimal depth is an expressibility threshold

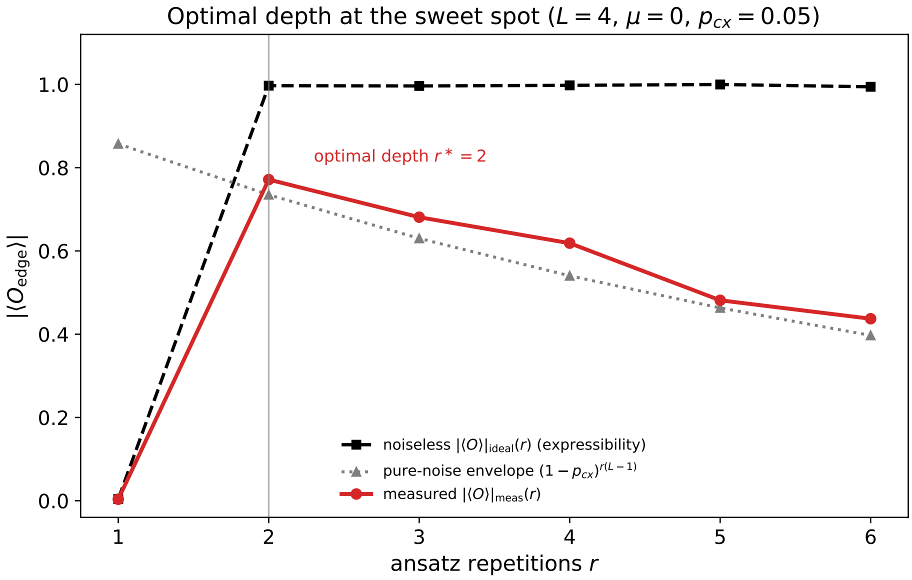

Figure 2: For $L = 4$, the noiseless edge string has an expressibility threshold: $r = 1$ fails, $r = 2$ succeeds, and deeper circuits mainly accumulate CNOT noise.

3.3 Noisy optimization and backend calibration

channel	strength	$\lvert\langle O_{\mathrm{edge}}\rangle\rvert$ noisy		fidelity to exact GS	
		fixed θ^*	noise-aware	fixed θ^*	noise-aware
depolarizing	0.08	0.502	0.501	0.998	1.000
amplitude damping	0.08	0.422	0.565	0.998	0.994
amplitude damping	0.12	0.266	0.337	0.998	0.966

Table 1: Best-case versus noise-aware optimization at $L = 4$, $r = 3$, $\mu = 0$. Under amplitude damping the measured edge string rises while the noiseless fidelity to the exact ground state falls at strong damping.

4 On the AI collaboration

References

[1] A. Yu. Kitaev. Unpaired Majorana fermions in quantum wires. *Physics-Uspekhi*, 44(10S):131–136, 2001. doi: 10.1070/1063-7869/44/10S/S29.